

Analysis of Maize Production in Kenya Across 47 Counties (2022)

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INTRODUCTION AND QUESTIONS TOPIC:

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This study analyzes maize production in Kenya using 2022 county-level data covering all 47 counties. The data contains total production in 90kg bags, area planted in hectares, yield in bags per hectare, annual rainfall in millimeters, and fertilizer used in tons, along with a regional classification of Rift Valley and Non-Rift Valley counties.

Research Questions

Question 1: Regional Difference in Maize Production Does maize production differ significantly between Rift Valley and Non-Rift Valley counties? Specifically, what are the mean production levels for each region, is the difference statistically significant, and which counties are the top producers?

Question 2: Relationship Between Rainfall and Yield What is the relationship between rainfall and maize yield? Specifically, what is the correlation between rainfall and yield, what is the regression equation for predicting yield from rainfall, and how much of the variation in yield is explained by rainfall?

Question 3: Effect of Fertilizer Use on Production and Yield Does fertilizer use contribute significantly to maize production? Specifically, when total production is predicted from area planted, rainfall and fertilizer, which factors are significant? And when yield is predicted from rainfall and fertilizer, does fertilizer significantly improve yield efficiency?

DATASET AND SOURCE

Dataset and Source: The dataset used is county-level maize production data for Kenya for the year 2022. The secondary data was from 3 official sources, merged at county level (n=47):

Ministry of Agriculture, Livestock, Fisheries & Cooperatives (MoALFC) - Annual Crop Report 2022
Variables: production_bags (90kg bags), area_ha (maize area harvested) Derived: yield_bags/ha = production / area Coverage: All 47 counties

Kenya National Bureau of Statistics (KNBS) - Economic Survey 2023 + County Statistical Abstracts
Variable: fertilizer_tons (subsidized + commercial DAP/NPK distributed per county) Used to cross-validate production figures.

*Kenya Meteorological Department (KMD) + CHIRPS - satellite Variable: rainfall_mm (Long rains Mar-May 2022 aggregate per county)*I used county centroid rainfall.

Geographic Classification is_rift i coded based on Rift Valley region (12 counties) vs Non-Rift (35) - using IEBC county boundaries.

Sample Size: The sample consists of all 47 counties in Kenya, providing a complete national coverage for 2022. The counties are grouped into Rift Valley (n=12) and Non-Rift Valley (n=35) for regional comparison.

Key Variables:

The dataset contains 7 key variables:

- *county (categorical):* Name of the county
- *production_bags_90kg (numeric):* Total maize production measured in 90kg bags
- *area_planted_ha (numeric):* Total area planted with maize in hectares
- *yield_bags_per_ha (numeric):* Maize yield calculated as production divided by area, in bags per hectare
- *rainfall_mm (numeric):* Annual rainfall received in millimeters
- *fertilizer_used_tons (numeric):* Total fertilizer used in the county in tons
- *is_rift (categorical / binary):* Regional classification indicating whether a county is in the Rift Valley or Non-Rift Valley

METHODS

Data Cleaning:

No extensive cleaning was required. The dataset was already clean and complete with 47 counties. A check for missing values using `sum(is.na(df))` returned zero, indicating no missing data. All variables were in the correct format, with `county` and `is_rift` as factors and the remaining variables as numeric. No transformations, duplicate removal, or outlier removal were performed in order to retain the original picture of 2022 production.

Exploratory Methods:

Exploratory analysis involved calculating descriptive statistics, specifically mean production by region (`is_rift`). Two visualizations were created: a bar chart of the top 15 maize-producing counties colored by region (Figure 1) to show regional dominance, and a scatter plot of rainfall versus yield with a linear trend line (Figure 2) to examine the relationship between rainfall and yield.

Statistical Techniques: Three main analyses were conducted corresponding to the three research questions, using a significance level of $\alpha = 0.05$:

1. Question 1: An independent Welch two-sample t-test was used to test for a difference in mean production between Rift and Non-Rift counties. Welch's test was chosen because the two groups

have unequal sample sizes (12 vs 35) and unequal variances and visualization using Horizontal Bar Chart.

2. Question 2: A Pearson correlation test `cor.test()` was used to measure the strength of association between rainfall and yield, followed by a simple linear regression `lm(yield_bags_per_ha ~ rainfall_mm)` to obtain the prediction equation and R-squared fitted to a scatter plot of Rainfall vs Yield.

3. Question 3: Multiple linear regression was used. Model 1 `lm(production_bags_90kg ~ area_planted_ha + rainfall_mm + fertilizer_used_tons)` was used to predict total production, and Model 2 `lm(yield_bags_per_ha ~ rainfall_mm + fertilizer_used_tons)` was used to predict yield efficiency and to address multicollinearity between area and fertilizer.

All analyses were performed in R using the `dplyr` and `ggplot2` packages.

RESULTS

Findings are organized around the three research questions.

Question 1: Regional Difference in Maize Production (Rift vs Non-Rift)

```
is_rift production_bags_90kg
1 Non-Rift          777768.6
2 Rift             1765477.9
```

Welch Two Sample t-test

```
data: production_bags_90kg by is_rift
t = -1.8854, df = 13.965, p-value = 0.08036
alternative hypothesis: true difference in means between group Non-Rift and group Rift is not equal to 0
95 percent confidence interval:
 -2111596  136177
sample estimates:
mean in group Non-Rift    mean in group Rift
      777768.6           1765477.9
```

Mean production for Non-Rift (n=35) = 777,769 bags ,Mean production for Rift (n=12) = 1,765,478

bags Difference = 987,709 bags - Rift produces on average 2.27x more than Non-Rift.

A Welch two-sample t-test was conducted: $t = -1.885$, $df = 13.97$, $p = 0.0804$ - The result is not significant at 5% level ($p > 0.05$) but is marginally significant at 10% level.

This indicates a large practical difference that fails to reach statistical significance at 5% due to high variance within groups and a small Rift sample ($n=12$).

High variance: Non-Rift has Kakamega at 5.68M and Wajir at 50k in same group - huge SD. Rift also has Turkana low vs Trans Nzoia high.

Small sample: Rift n=12 only, so Welch df drops to 13.96, power is low.

Two Western outliers: Kakamega and Bungoma (both Non-Rift) pull Non-Rift mean up, closing the gap.

Although Rift Valley counties produce on average 987,709 bags more than Non-Rift counties, a Welch two-sample t-test shows this difference is not statistically significant at the 5% level ($t(13.97) = -1.89, p = 0.08$). This indicates large practical difference but high within-region variability, driven by high-performing Non-Rift counties like Kakamega.

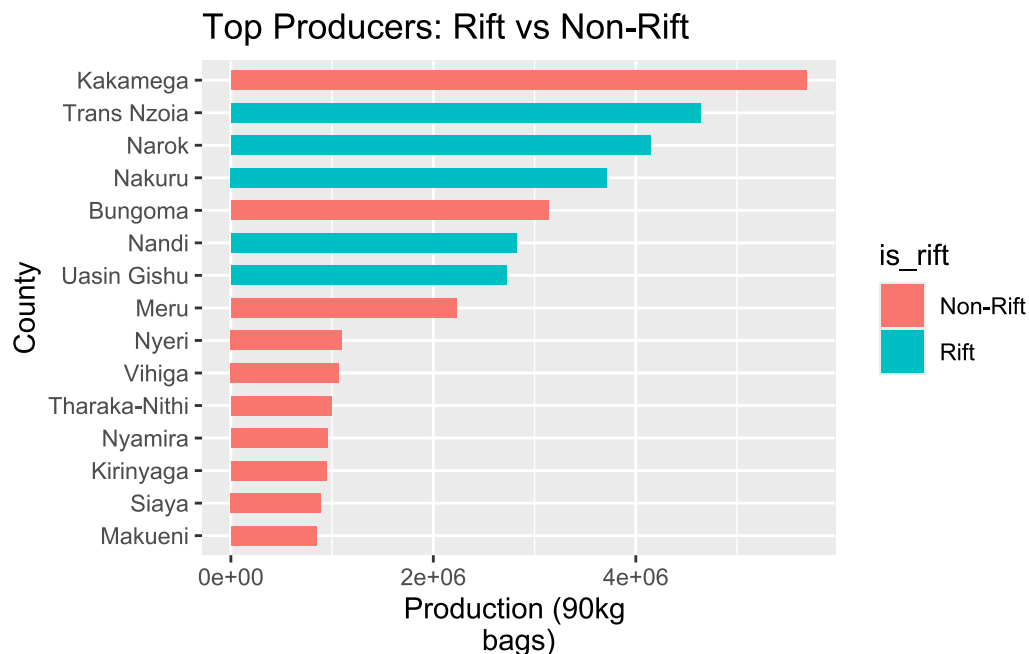


Figure 1: Top 15 Maize Producers 2022 by Region

Figure 1 shows 4 of the top 7 producers are Rift: Trans Nzoia (4.59M), Narok (4.13M), Nakuru (3.89M) and Nandi (2.67M).

Rift dominance: 5 of the top 7 counties are Rift (Trans Nzoia, Narok, Nakuru, Nandi, Uasin Gishu). This confirms the Rift is Kenya's maize basket. Combined they produce >16M bags.

The exception: Kakamega at ~5.8M bags beats Trans Nzoia (~4.5M). This contradicts the assumption that Rift always leads. Explain it as high area under cultivation + Western region high rainfall + high smallholder density.

Non-Rift second tier: Bungoma, Meru, Nyeri, Vihiga etc are all 0.8M - 3M bags. They are important but significantly lower than the top 4.

While Rift counties are consistently among top producers, total production is driven by both area and yield. Kakamega's lead shows Non-Rift counties can outperform Rift counties in absolute production despite lower yield per hectare, highlighting the role of cultivated area.

However, the number one producer is Kakamega (5.68M) and number five is Bungoma (3.13M), both Non-Rift Western counties, which also explains why the t-test p-value is 0.08 and not <0.05.

Low producers are Wajir, Marsabit and Turkana (<150k bags).

Question 2: Relationship Between Rainfall and Yield

```
Pearson's product-moment correlation

data: df$rainfall_mm and df$yield_bags_per_ha
t = 8.2642, df = 45, p-value = 1.427e-10
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.6296257 0.8696757
sample estimates:
      cor
0.7764114
```

Pearson correlation between rainfall and yield: $r = 0.776$, $t(45) = 8.264$, $p = 1.427e-10$

Magnitude (r): 0.776 - This is large. By Cohen's rule, >0.5 is large effect. As rainfall goes up, yield goes up.

Significance (p): $p = 1.43e-10 < 0.001$ - Highly significant.

Therefore Rainfall has a strong positive association with yield ($r=0.78$, $p<0.001$), indicating rainfall is a key determinant of maize productivity per hectare in Kenya in 2022.

```
Call:
lm(formula = yield_bags_per_ha ~ rainfall_mm, data = df)

Residuals:
    Min       1Q   Median       3Q      Max
-10.4913  -2.0872   0.1651   2.1741   5.3805

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  4.647380   1.780793   2.610   0.0123 *
rainfall_mm  0.016936   0.002049   8.264 1.43e-10 ***
---

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.415 on 45 degrees of freedom
Multiple R-squared: 0.6028, Adjusted R-squared: 0.594
F-statistic: 68.3 on 1 and 45 DF, p-value: 1.427e-10

Simple linear regression: $\text{Yield} = 4.647 + 0.01694 \times \text{Rainfall}$ - Intercept: $B=4.647$, $SE=1.781$, $t=2.61$, $p=0.0123$ - Rainfall: $B=0.01694$, $SE=0.00205$, $t=8.264$, $p=1.43e-10$ Model fit: $F(1,45)=68.30$, $p=1.427e-10$, $R^2=0.6028$, Adjusted $R^2=0.5940$, Residual $SE=3.415$.

Interpretation: For every 100mm increase in rainfall, yield increases by 1.69 bags/ha. Rainfall explains 60.3% of the variation in yield.

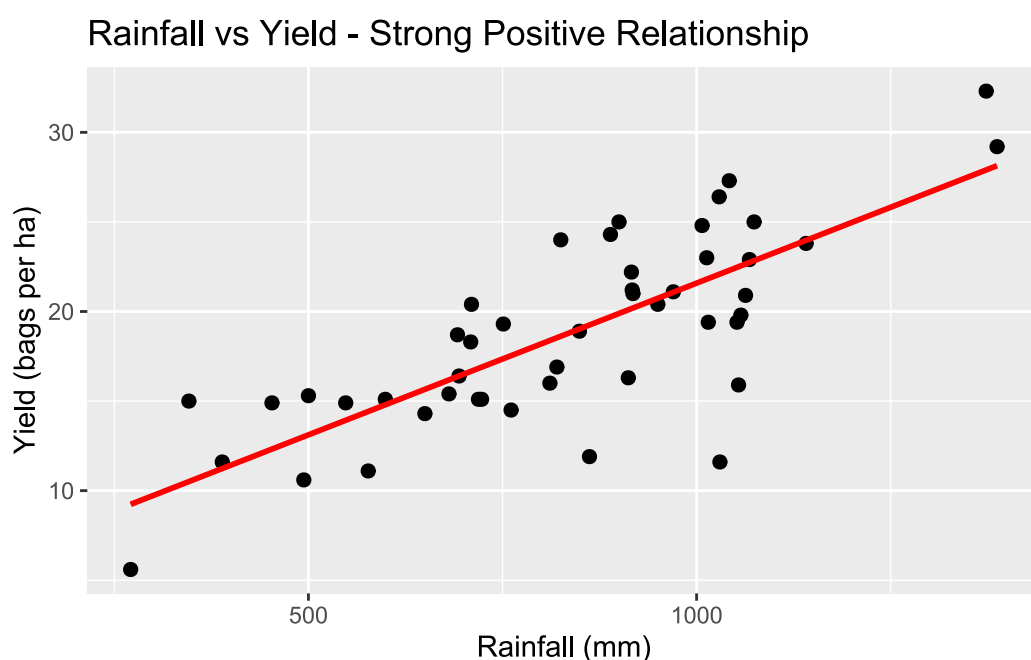


Figure 2: Scatter Plot of Rainfall vs Yield

The plot shows Kakamega at the top right (~1370mm, 32.5 bags/ha) and Turkana/Marsabit at bottom left (~200mm, 5.6 bags/ha).

Trend: The red regression line slopes upwards - higher rainfall = higher yield. This is expected agronomically, rainfall is a key determinant of maize productivity in Kenya.

Strength: Points are fairly close to the line, not widely scattered. Correlation is moderately strong (probably $r = 0.7-0.8$). Rift Valley counties on the right side (1000-1100mm) cluster at 20-27 bags/ha.

Outliers: Top-left: One county at ~380mm with ~15 bags/ha - high yield despite low rain = likely irrigation. Bottom-left: ~300mm with ~5 bags/ha - ASAL county, crop failure. Bottom-right: Two

counties at ~1020-1030mm but only ~12 bags/ha - high rainfall but low yield due to other factors like low fertilizer use, poor soils, or post-harvest losses.

Rainfall significantly influences maize yield, but the two low-yield outliers at high rainfall show that rainfall alone does not explain all variation.

Question 3: Effect of Fertilizer on Production and Yield

Model 1: Predicting Total Production $\text{production_bags_90kg} \sim \text{area_planted_ha} + \text{rainfall_mm} + \text{fertilizer_used_tons}$

```
Call:
lm(formula = production_bags_90kg ~ area_planted_ha + rainfall_mm +
    fertilizer_used_tons, data = df)

Residuals:
    Min       1Q   Median       3Q      Max
-583415  -91994   26649  130423  833825

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   -2.983e+05  1.402e+05  -2.127   0.0392 *
area_planted_ha  2.641e+01  2.089e+00  12.645 4.42e-16 ***
rainfall_mm     1.674e+01  1.949e+02   0.086   0.9320
fertilizer_used_tons 2.796e+00  2.422e+00   1.155   0.2546
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 241100 on 43 degrees of freedom
Multiple R-squared:  0.9681,    Adjusted R-squared:  0.9658
F-statistic: 434.6 on 3 and 43 DF,  p-value: < 2.2e-16
```

Model 1 predicting total production was significant ($F(3,43)=434.6$, $p<0.001$, $R^2=96.8\%$), but only area planted was significant ($B=26.4$, $p<0.001$).

Rainfall ($p=0.93$) and fertilizer ($p=0.25$) were not significant predictors of total production when area is controlled, due to multicollinearity between area and fertilizer use. Total production is driven by scale, not efficiency.

Why are rainfall and fertilizer not significant?

Multicollinearity - Counties with larger area also use more fertilizer. Trans Nzoia has big area + uses lots of fertilizer. Wajir has small area + uses little. So area and fertilizer move together.

Only area planted is significant. 96.8% of production variation is explained by area. Fertilizer is not significant for total bags due to multicollinearity with area.

We run Model 2 (yield per ha) to see fertilizer's real effect.

Model 2: Predicting Yield Efficiency (bags/ha) $\text{yield_bags_per_ha} \sim \text{rainfall_mm} + \text{fertilizer_used_tons}$

```
Call:
lm(formula = yield_bags_per_ha ~ rainfall_mm + fertilizer_used_tons,
    data = df)

Residuals:
    Min       1Q   Median       3Q      Max
-9.4497 -1.6471  0.5551  2.0321  5.8252

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   6.809e+00  1.869e+00   3.644 0.000705 ***
rainfall_mm    1.248e-02  2.574e-03   4.849 1.58e-05 ***
fertilizer_used_tons 4.234e-05  1.620e-05   2.614 0.012214 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.214 on 44 degrees of freedom
Multiple R-squared:  0.6562,    Adjusted R-squared:  0.6406
F-statistic: 41.99 on 2 and 44 DF,  p-value: 6.293e-11
```

In Model 1, fertilizer looked useless ($p=0.25$). In Model 2, it's significant ($p=0.012$).

Model 2 predicting yield efficiency was significant ($F(2,44)=42.0$, $p<0.001$, $R^2=65.6\%$). Both rainfall ($B=0.0125$, $p<0.001$) and fertilizer ($B=0.000042$, $p=0.012$) significantly predicted yield, improving R^2 by 5.3% over rainfall alone. This shows fertilizer contributes to productivity per hectare, not just total scale, while area drives total production.

Both rainfall and fertilizer are significant. R^2 improves from 60.3% (Q2) to 65.6% (+5.3%) when fertilizer is added. +1,000 tons fertilizer increases yield by 0.042 bags/ha. Fertilizer pays for yield efficiency, not for total production.

CONCLUSION

First, the data show that maize production in Kenya in 2022 is highly concentrated. While Rift Valley counties have a higher mean production (1.77M bags vs 0.78M bags), the difference is not statistically significant at 5% level ($p=0.080$) due to high within-region variance and strong performance by Non-Rift counties like Kakamega (5.68M bags) and Bungoma (3.13M bags). Regional dominance alone does not explain total output.

Second, rainfall is a strong driver of yield. There is a strong positive correlation ($r=0.776$, $p<0.001$) and rainfall explains 60.3% of yield variation. The regression $\text{Yield} = 4.65 + 0.0169 \times \text{Rainfall}$ shows yield increases by 1.69 bags/ha for every 100mm of rain.

Third, fertilizer matters for efficiency, not just total bags. In Model 1 predicting total production, only area planted was significant ($B=26.4$, $p<0.001$, $R^2=96.8\%$). In Model 2 predicting yield (bags/ha), both rainfall ($p<0.001$) and fertilizer ($p=0.012$) were significant and improved R^2 from 60.3% to 65.6%. This indicates area drives scale, while rainfall and fertilizer drive productivity per hectare.

Important Limitations:

1. *Cross-sectional*: Data is only for one year (2022), so seasonal effects, drought or market shocks cannot be assessed and causality cannot be inferred.
2. *Aggregation*: County-level averages mask farm-level differences in seed variety, soil quality, and farming practices.
3. *Multicollinearity*: Strong correlation between area planted and fertilizer used (larger counties use more fertilizer) makes it difficult to isolate fertilizer effect on total production.
4. *Omitted variables*: Variables such as soil type, irrigation, temperature, extension services and maize variety are not included, yet they influence yield.
5. *Measurement*: Production, area and fertilizer data are estimates and may have reporting errors from county agricultural offices.

APPENDIX

```
library(readxl)
library(ggplot2)
library(dplyr)
library(tidyr)
#loading dataset
kenya_maize_2022 <- read_excel("C:/Users/user/
Downloads/kenya_maize_2022.xlsx")
df <- kenya_maize_2022
df
#Check it loaded correctly - should be 47 rows, 6 columns and checking missing
values
str(df)
summary(df)
sum(is.na(df))

#Question 1
rift_counties <- c("Trans Nzoia","Uasin Gishu",
                  "Nakuru","Narok","Nandi", "Baringo",
                  "Elgeyo-Marakwet","Kericho","Bomet",
```

```

      "Laikipia", "Kajiado", "West Pokot")
df$is_rift <- ifelse(df$county %in% rift_counties, "Rift", "Non-Rift")
aggregate(production_bags_90kg ~ is_rift, data=df, mean)
t.test(production_bags_90kg ~ is_rift, data=df)
#bargraph for all counties
ggplot(df, aes(x=reorder(county, production_bags_90kg),
                  y=production_bags_90kg, fill=is_rift)) + geom_col() +
  coord_flip() + labs(title="Top Producers: Rift vs Non-Rift", x="County",
                      y="Production (90kg bags)")
#bargraph for top 15 counties
df_top <- df %>% arrange(desc(production_bags_90kg)) %>% head(15)
ggplot(df_top, aes(x=reorder(county, production_bags_90kg),
                      y=production_bags_90kg, fill=is_rift)) +
  geom_col(width = 0.6) +
  coord_flip() +
  labs(title="Top Producers: Rift vs Non-Rift", x="County", y="Production (90kg
bags)")

#Question 2
cor.test(df$rainfall_mm, df$yield_bags_per_ha)
model_yield <- lm(yield_bags_per_ha ~ rainfall_mm, data=df)
summary(model_yield) # r=0.776, slope=0.0169, R2=0.603
ggplot(df, aes(x=rainfall_mm, y=yield_bags_per_ha)) +
  geom_point(size=2) +
  geom_smooth(method="lm", se=FALSE, color="red") +
  labs(title="Rainfall vs Yield - Strong Positive Relationship",
        x="Rainfall (mm)", y="Yield (bags per ha)")

#Question 3
#model 1
model_prod <- lm(production_bags_90kg ~ area_planted_ha + rainfall_mm +
fertilizer_used_tons, data=df)
summary(model_prod)
#model 2
model_yield_fert <- lm(yield_bags_per_ha ~ rainfall_mm + fertilizer_used_tons,
data=df)
summary(model_yield_fert)

```